Marwadi University	**Marwadi University** **Faculty of Technology** **Department of Information and Communication Technology**	
Subject:	**Aim :**	
Open handed problem 1	**Date:15-07-2026**	**Enrolment No: 92400133131**

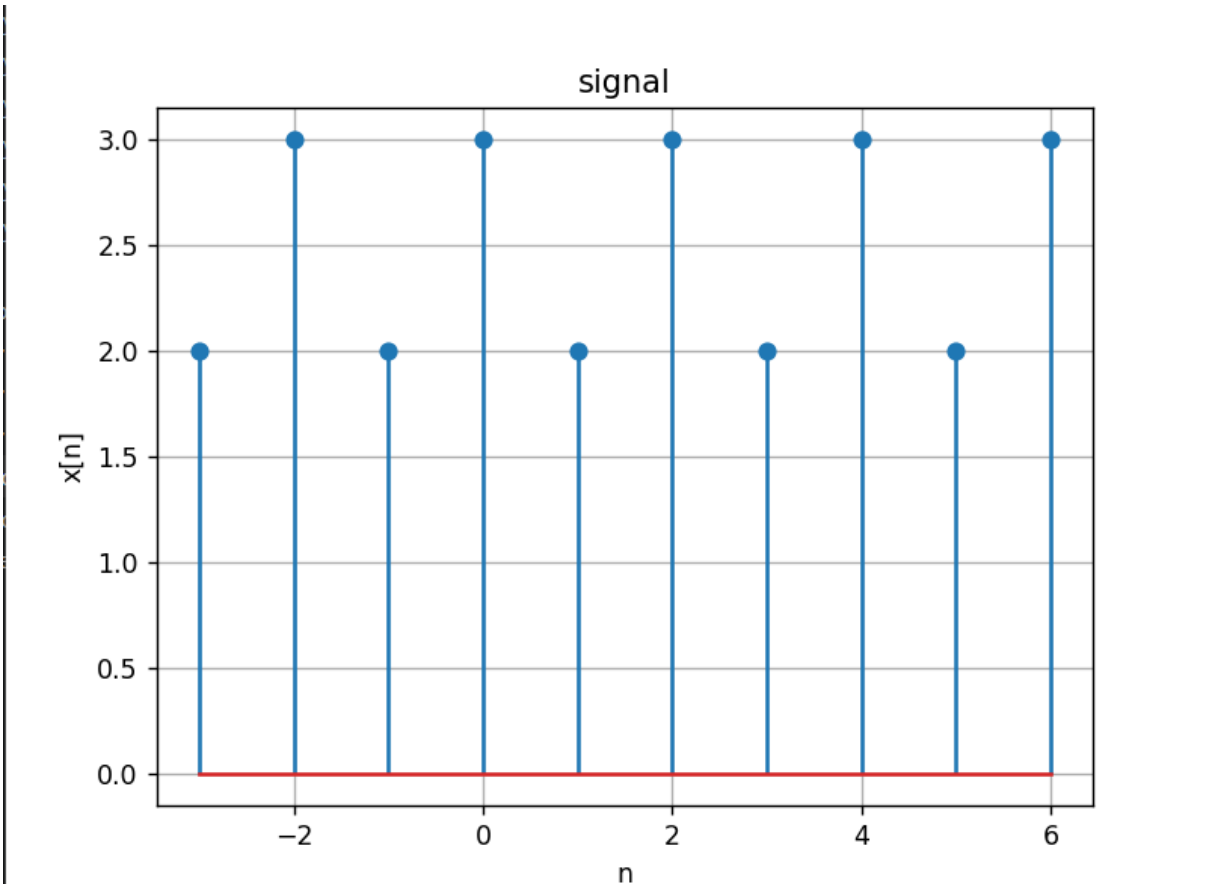
Code 1 :-

```python
import numpy as np
import matplotlib.pyplot as plt

n = np.arange(-3, 7)
x = np.where(n % 2 == 0, 3, 2)

plt.stem(n, x)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



Code :-

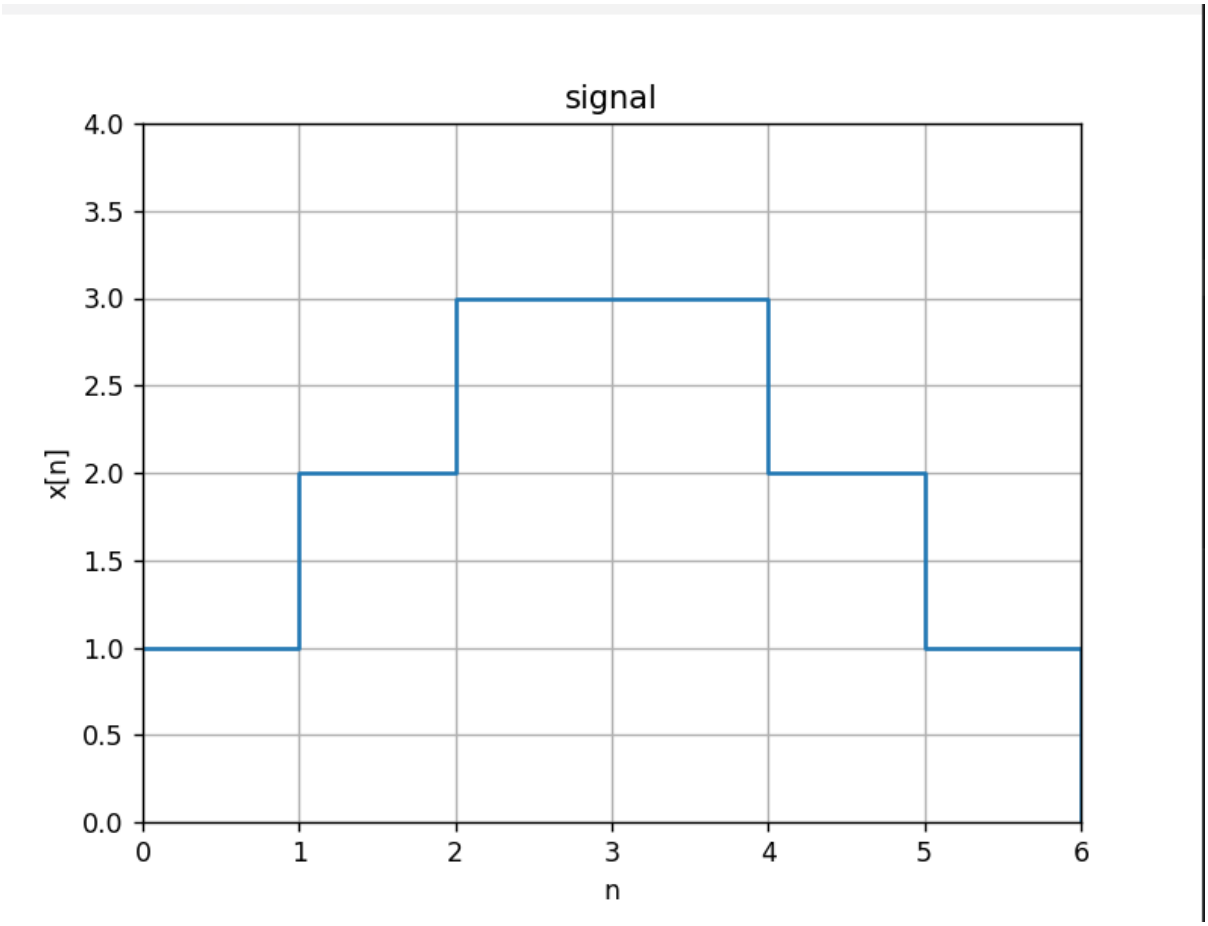
```python
import numpy as np
import matplotlib.pyplot as plt
```

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```
n = np.arange(0, 7)
x = [1, 2, 3, 3, 2, 1, 0]

plt.step(n, x, where="post")
plt.xlim(0, 6)
plt.ylim(0, 4)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



Code 3 :- `import numpy as np`

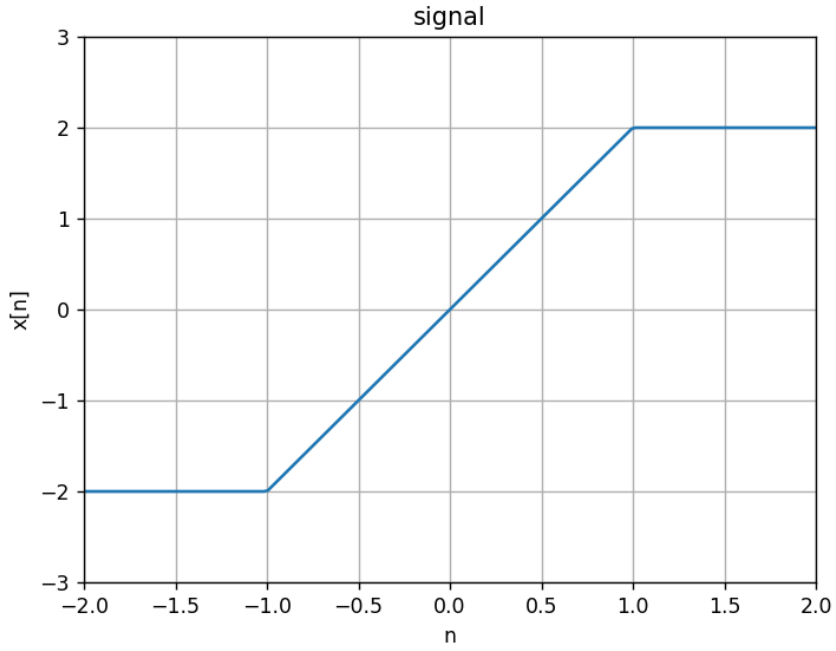
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```
import matplotlib.pyplot as plt

n = np.linspace(-2, 2, 200)
x = np.piecewise(n,
                 [n < -1, (-1 <= n) & (n <= 1), n > 1],
                 [-2, lambda n: 2*n, 2])

plt.plot(n, x)
plt.xlim(-2, 2)
plt.ylim(-3, 3)
plt.title("signal")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



Code 4 :-
```
import numpy as np
import matplotlib.pyplot as plt
```

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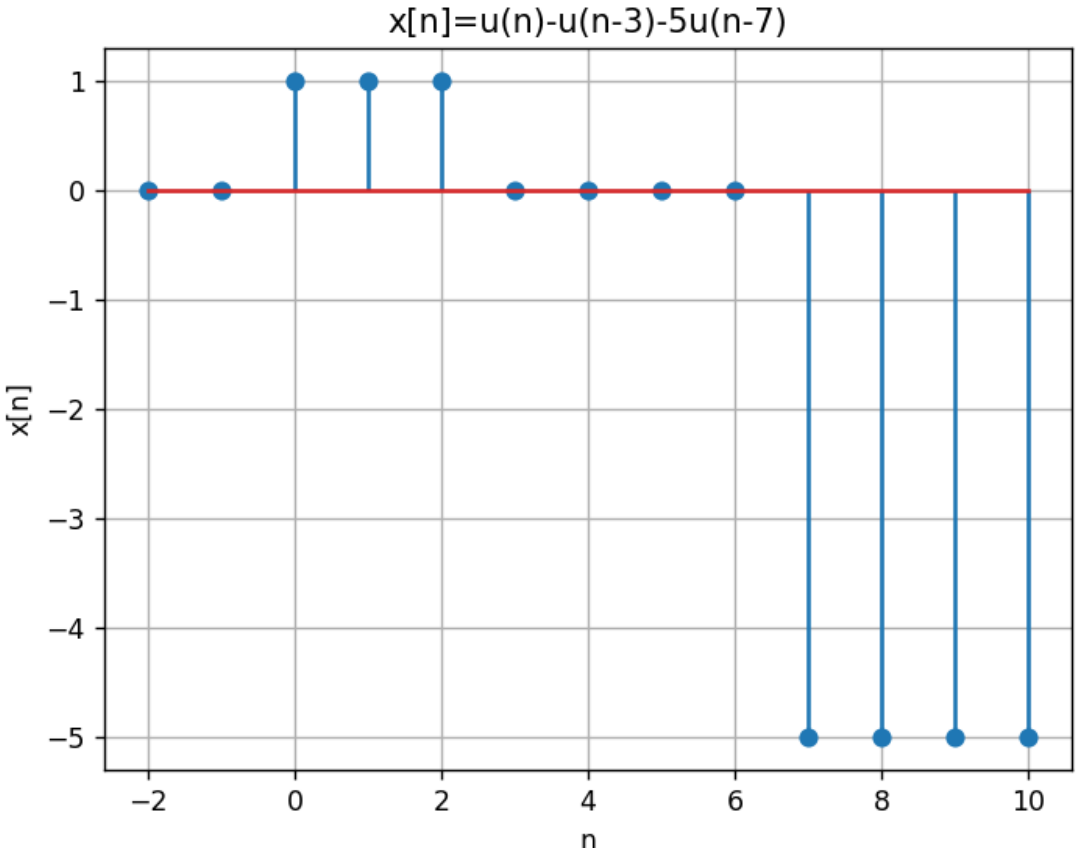
```
n = np.arange(-2, 11)

u = lambda x: (x >= 0).astype(int)

x = u(n) - u(n-3) - 5*u(n-7)

plt.stem(n, x)
plt.title("x[n]=u(n)-u(n-3)-5u(n-7)")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



Code 5:- `import numpy as np`

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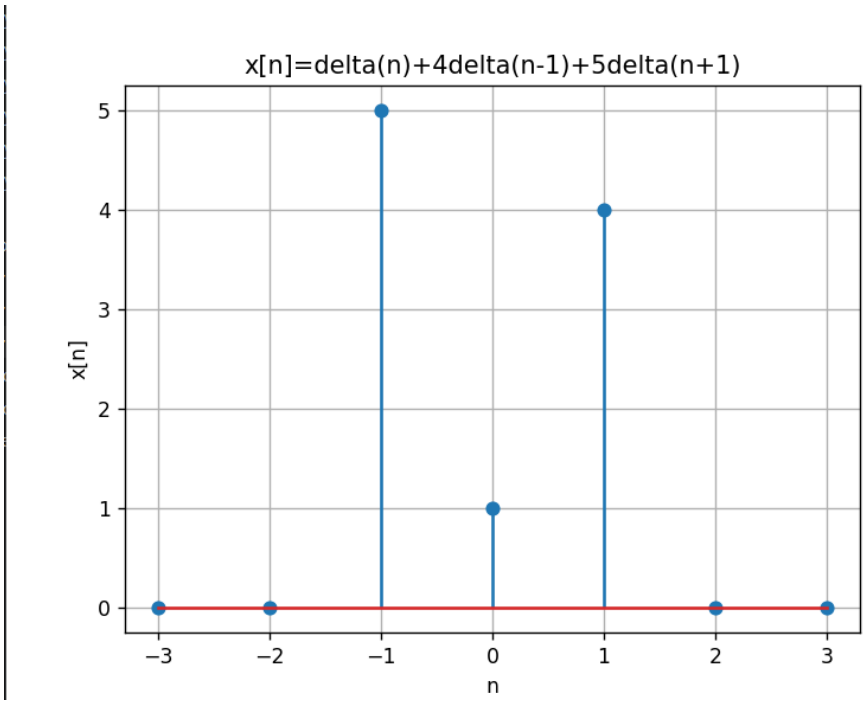
```python
import matplotlib.pyplot as plt

n = np.arange(-3, 4)

x = np.zeros(len(n))
x[n == 0] = 1
x[n == 1] = 4
x[n == -1] = 5

plt.stem(n, x)
plt.title("x[n]=delta(n)+4delta(n-1)+5delta(n+1)")
plt.xlabel("n")
plt.ylabel("x[n]")
plt.grid()
plt.show()
```

Output :-



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Github link :- https://github.com/farhan-web404/Digital-Signal-and-Image-Processing/tree/main/lab%201